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Alkali-Enhanced Mechanical Durability of Polymers in Porous Media: Insights from Core Flooding Experiments

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Abstract

Adsorption of alkali on rock surfaces is a well-established phenomenon in high Total Acid Number (TAN) oil reservoirs, particularly noticeable during Alkali Polymer (AP) flooding. Despite this, the influence of alkali on the mechanical degradation of polymers in these environments remains insufficiently understood. This study explores the role of alkali in enhancing the mechanical stability of polymers under shear stress conditions prevalent in porous media.

Utilizing polyacrylamide polymer products supplied by two vendors, alkali-polymer slugs were prepared for this work. Core flooding experiments were conducted at injection velocities representative of field-scale operations to induce and study the mechanical degradation of polymer chains. These experiments were enabled assessing the rate of polymer mechanical degradation, for which viscosity measurements served as a primary metric. To encompass a range of field scenarios, injection velocities varied from 1 to 20 feet per day. Additionally, core plugs with both low and high permeability were employed to replicate the diverse shear rates encountered in heterogeneous reservoirs.

Our findings demonstrate that the presence of alkali clearly enhances the mechanical stability of polymers. This improvement is likely attributed to the increased anionicity caused by alkali. Another contributor could be the role of alkali as changing rock surfaces to be positively charged, which seems to reduce polymer adsorption, thereby helping to maintain viscosity control. Core plugs with lower permeability exhibited higher shear rates, leading to a marginally greater breakdown of polymer chains and consequently more significant viscosity reduction in effluents. As anticipated, higher injection velocities, which simulate the conditions near wells, resulted in greater viscosity losses. In comparing the two polymer products, Flopaam from vendor-B showed superior viscosity control, whereas the vendor-A product exhibited enhanced injectivity. The injectivity outcomes for core floods in presence of crude oil (two-phase) align with those from single-phase core floods when injecting polymer/alkaline polymer (AP) slugs. The enhanced recovery factor attributed to the AP slug injection (23% vs 2%) is confirmed to result not only from the in-situ saponification process but also from the increased mechanical stability of the AP slug.

This study not only sheds light on the flow dynamics of polyacrylamide polymers but also underscores the importance of mechanical stability in situ. Such stability is a critical factor affecting both injectivity

and mobility in porous media. Therefore, it is imperative to consider these findings as part of the screening criteria for field applications, especially where AP flooding is employed. This research thus provides a pivotal step forward in understanding and optimizing polymer-based enhanced oil recovery techniques in challenging reservoir conditions.

Introduction

The chemical enhanced oil recovery (cEOR) processes that use polymer and alkali-polymer (AP) have attracted significant attention due to their ability to improve the efficiency of oil production (Gbadamosi et al., 2022). Polymers increase the viscosity of displacing fluids and improve the sweep efficiency (Mahajan et al., 2021; Mohsenatabar Firozjahi and Saghafi, 2020; Scott et al., 2020). AP slug results in in-situ saponification production which can lead to an optimal interfacial tension reduction and reduces capillary pressure in addition to altering the viscosity of injected fluids (Ding et al., 2020; Hincapie et al., 2022c).

Polymers demonstrate significant mechanical degradation when subjected to harsh reservoir conditions, which may limit their injectivity and overall efficacy (Al-Shakry et al., 2018). Mechanical degradation causes the viscosity loss of displacing fluid and hence reduce the mobility ratio. Similarly, alkali adsorption to rock surface is a major challenge and negatively impacts the chemical slug efficacy owing to pH-front delay (Tagavifar et al., 2018). Alkali is sometimes added to complex formulations of alkali-surfactant (AS)/alkali-surfactant- polymer (ASP) to act as a sacrificing agent and reduces surfactant adsorption by neutralizing reactive cationic sites on the rock surface (Hazarika and Gogoi, 2020). This is expected to enhance the efficiency of the chemical slug in recovering additional oil. Furthermore, alkali can also induce geochemical interactions under high pH conditions and result in scale precipitation (Fernandes and Broekmans, 2013). Research indicates that the compatibility of alkali types greatly influences the chemical behavior of EOR formulations (Hassan et al., 2022). NaOH being more aggressive in nature has caused pilot project failures due to significant pH value increase and activating silica scale formation (Hunter et al., 2013; McInnis et al., 2013). However, Na_2CO_3 being less aggressive and having better control of pH increase has shown potential to be considered for cEOR pilot projects (Nurmi et al., 2023; Schumi et al., 2020; Tahir et al., 2024b).

Injection of an AP slug leads to better oil mobilization compared to either using alkali or polymer alone (Ahmed et al., 2023). Alkali reduces the residual oil saturation by wettability alteration and reducing the interfacial tension and polymer enhances the sweep efficiency due to better mobility control. However, challenges arising from combining alkali and polymer into more complex formulations impact the treatment of backproduced fluids, complex slug design, stability of chemical slugs and cause formation damage (Ahmed et al., 2023; Chang et al., 2024). Specific alkaline formulations showed better performance in high temperature reservoirs and high salinity, where traditional polymers solutions usually fail (Davarpanah and Mirshekari, 2020). Therefore, understanding the interaction between alkali and polymers under various reservoir conditions is critical to maximize oil recovery (Gbadamosi et al., 2022; Scott et al., 2020).

The role of alkali in increasing mechanical durability of polymers is critical to optimizing the enhanced processes of cEOR. By adjusting pH levels, alkali improves polymer solubility and increases polymer viscosity, essential for improving the characteristics of the fluid flowing through porous medium (Olajire, 2014). Similarly, aging of AP slugs results in viscosity enhancement of a chemical slug in-situ during slug propagation from injector to producer (Hincapie et al., 2022b; Nurmi et al., 2023; Tahir et al., 2024a). Followingly, addition of alkali can mitigate polymer degradation by promoting the formation of stronger polymer networks that can support shear forces during injection (Agi et al., 2018). One contributing factor to mitigating polymer degradation is the preferred alkali modifying the rock surface, which keeps the polymer slug concentration high enough to hold a good viscosity. Second, a higher pH-value of the AP slug is expected to result in stronger repulsive forces among polymer chains leading to higher viscosity. Despite some indications that alkali treatment may enhance mechanical stability, there is a notable lack of

comprehensive literature supporting this claim, highlighting the need for further experimental investigation to validate its effects.

In this study, we conducted both single-phase and two-phase core flooding experiments to test the hypothesis that the presence of alkali enhances polymer stability against mechanical degradation. The observed incremental oil recovery from the alkali-polymer (AP) slug is attributed not only to improved fluid-fluid (f-f) interactions but also to enhanced sweep efficiency, resulting from the increased mechanical stability of the polymer solution.

Approach

This study focuses on fluid-rock-fluid interactions and viscosity measurements, employing the following approaches:

- **Fluid-fluid (f-f) interactions**, with an emphasis on interfacial tension (IFT) measurements and phase behavior tests.
- **Single-phase core floods** using polymer and alkali-polymer (AP) solutions to simulate slug injectivity.
- **Viscosity measurements** of both fresh solutions and effluents to assess the mechanical stability of the chemical slug.
- **Two-phase core floods** conducted at realistic injection velocities (1 ft/day) to evaluate mechanical stability under field-scale conditions.

Fluids and Rock Types

Reservoir Information: Experiments are conducted under reservoir conditions of pressure and temperature (8th Torton Horizon) of the Matzen field. The reservoir consists of clastic sandstone with a permeability range from 150 mD to several Darcies, indicating significant heterogeneity. The average porosity ranges from 28% to 30% and the producing formation has a thickness of 5 meters. The reservoir temperature at the production well is 49 °C; however, a lower temperature is expected at the injection well due to the cooling effect. The live reservoir oil has a viscosity of 20 mPa·s under reservoir conditions. The formation brine is a low-salinity brine with a measured salinity of 15.84 g/L.

Oil Properties: Crude oil from the S-85 well is used for the experimental study. The crude oil is moderately degraded and reacts with alkali and results in soap generation (saponification). SARA analysis of the crude oil sample showed that it contains 42% aromatic compounds, 39% saturated hydrocarbons, 16% resins, and 3% asphaltenes. The measured total acid number (TAN) of the crude oil is 2.14 mg KOH/g and the saponifiable acid content is 41 g/mol/g (using 7.7 g/L Na₂CO₃). The viscosity of the dead oil at reservoir temperature is 56 mPa·s and its density is 0.891 g/cm³. Fluid-fluid (f-f) interactions are performed using dead crude oil. However, 11 wt% cyclohexane is added to the dead crude oil to match the viscosity of live oil (20 mPa·s) at the reservoir temperature of 49 °C.

Synthetic Brines: Two types of synthetic brines are prepared for this study as shown in Table 1. "8 TH WTP" represents the hard brine (containing Ca²⁺ and Mg²⁺ compounds) and is a simplified version of the formation brine. This brine is used to initialize the core plug saturations as well as for brine flooding and polymer slug preparation. "Soft 8 TH WTP" is defined as the second brine without Ca²⁺ and Mg²⁺ compounds. This brine is used for the preparation of the AP slug as alkali poses a potential risk of inorganic scale precipitation due to its reactivity with Ca²⁺ and Mg²⁺ compounds present in the hard brine.

Table 1—Brine compositions used for this study

Salt	8 TH WTP (g/L)	Soft 8 TH WTP (g/L)
NaCl	22.48	22.62
KCl	0.16	0.16
MgCl ₂ *6H ₂ O	0.63	-
CaCl ₂ *2H ₂ O	0.94	-
NaHCO ₃	-	1.52
TDS	15.84	15.02

Polymers: Two polymer products were selected for this study for comparison purposes,. The target reservoir conditions were shared with two vendors and one product from each vendor was proposed for laboratory testing. Both polymers are high molecular weight hydrolyzed polyacrylamide (HPAM) products with reported molecular weights of 18-20 MDa for polymer product 1 (referred to as PP1 in the following) from Vendor A and 20-24 MDa for polymer product 2 (PP2) from Vendor B. Polymer solutions were prepared in 8 TH WTP brine to achieve a target viscosity of 20 mPa·s at a shear rate of 7.94 s⁻¹ to establish the comparison. Two temperatures—25 °C and 49 °C—were selected to represent the lower reservoir temperature near the injection well and the higher temperature deeper in the reservoir. The injection well has experienced prolonged water/polymer injection and is expected to have cooled the near-wellbore region. This is relevant because injectivity becomes more challenging at lower temperatures.

Alkali-polymer Solution: Washing soda, chemically known as sodium carbonate (Na₂CO₃), is used as an alkali to prepare AP solutions for this study. It is a low-cost chemical that helps prevent the risk of silica scale precipitation in production wells due to its less aggressive interaction with reservoir silicates (Nurmi et al., 2023). Initially, polymer solutions are prepared in soft 8 TH WTP brine at the same concentration as the polymer slug described in the previous section, to achieve the target viscosity. AP slugs are then prepared in-house by mixing Na₂CO₃ into the polymer solution using a magnetic plate stirrer. Thus, while polymer solutions are prepared in hard brine, AP solutions are prepared in soft brine, both using the same polymer concentration. The viscosity of the AP slugs is measured to cross- validate the polymer slug viscosity at similar concentrations for comparison and quality control. The selected alkali concentration (7,500 ppm) is based on laboratory testing from previous studies (Hincapie et al., 2022b; Schumi et al., 2020; Tahir et al., 2024b). This concentration increases the pH value while keeping it below 10.5, as a higher pH may trigger scale precipitation under the target reservoir conditions.

Core Plugs: Two types of Berea sandstone core plugs, with permeabilities of 100 mD and 300 mD, were used for single-phase core flooding to address reservoir heterogeneity. The lower permeability cores were selected to investigate the mechanical degradation caused by higher in-situ shear rates. Smaller core plugs (7 cm in length) were used for single-phase core flooding, while longer plugs (30 cm) were employed for two-phase core floods to mitigate capillary end effects and minimize dead volume errors. Two-phase core floods were conducted exclusively on the 300 mD core plugs. Computed tomography (CT) scans of the high-permeability (300 mD) sister core plugs revealed inhomogeneities, as reported by Scheurer et al (Scheurer et al., 2022). Additionally, the presence of clay and feldspar in the pore walls of these core plugs was documented by (Saleh, 2020; Saleh et al., 2021).

Methods

Chemical Slug Preparation and Viscosity Measurements

Slugs of both polymer products are prepared in 8 TH WTP brine. A mother solution with a concentration of 5,000 ppm is first prepared using an overhead stirrer at a high speed of 650 rpm for 15 minutes, followed

by a reduced speed of 335 rpm for 145 minutes. Diluted solutions at the desired concentrations are then prepared by mixing the mother solution with brine using a magnetic plate stirrer operating at 700 rpm for 130 minutes. Viscosity measurements of the diluted solutions are performed using a Kinexus Pro+ rheometer at two temperatures: 25 °C and 49 °C. Two polymer slug concentrations are selected to achieve a target viscosity of 20 mPa·s at a shear rate of 7.944 s⁻¹ at both temperatures. Based on the reported molecular weights of the two products, it is expected that PP1 will require a slightly higher concentration to reach the target viscosity.

AP solutions are also prepared in-house using a similar procedure to that used for polymer slug preparation. Mother and diluted solutions of polymer slugs are prepared in soft 8 TH WTP brine. A 7500 ppm concentration of alkali is slowly added at the end of the polymer dilution process. Typically it takes 5-10 minutes to dissolve the target alkali concentration. The viscosity of the prepared AP slug is measured using the Kinexus Pro+ rheometer. The polymer concentration in the AP slug matches that used to achieve the target viscosity of 20 mPa·s at a shear rate of 7.944 s⁻¹ at both temperatures. Due to the use of soft brine, the viscosity of the AP slug is expected to be slightly higher than that of the polymer slug at the same concentration. Viscosity measurements for both slugs are conducted over a shear rate range of 1-20 s⁻¹ at both temperatures. These measurements are performed prior to core flooding to establish baseline values for quality control and cross-validation.

Fluid-Fluid Interactions

Interfacial tension (IFT) was measured using a Kruss spinning drop tensiometer, which is capable of detecting very low IFT values (as low as 10⁻³ mN/m). An oil drop was introduced into the AP solution at room temperature, and a suitable rotation speed (3,000-7,000 rpm) was applied to maintain the elongated shape of the oil drop within the aqueous phase. Initial IFT measurements were taken at room temperature, as the system was prepared under ambient conditions. The heating jacket was then activated and set to 49 °C to measure IFT at reservoir temperature, ensuring the oil drop remained elongated. Three measurements were performed using fresh fluids to ensure reproducibility and minimize error. The results were averaged, and the standard deviation was calculated to report the final values.

Phase behavior tests were carried out in 10 mL glass pipettes sealed at the bottom. Each pipette was filled with 5 mL of the AP slug, followed by 5 mL of crude oil. The pipettes were then sealed at the top using a methane-oxygen flame and rotated for 48 hours at 50 rpm using a rotator. This rotation facilitated cross-flow between the fluids and promoted emulsion formation due to chemical interactions between the AP slug and polar compounds in the crude oil. After rotation, the pipettes were stored in an oven at 49 °C for a specified period. Emulsion stability was evaluated by visually analyzing changes in emulsion volume. Three pipettes were prepared for each formulation to ensure reproducibility and improve data quality. The average emulsion volume and standard deviation were calculated to report the final results.

Rock-Fluid Interactions

Single-Phase Core Flooding. *Injectivity Test:* Injectivity tests are conducted using two types of sandstone core plugs—300 mD and 100 mD—to investigate the impact of reservoir heterogeneity on mechanical stability. The experiments are performed using a custom-built setup shown in Fig. 1, with a Hassler cell serving as the core holder. Smaller core plugs, 7 cm in length, are used for single-phase injectivity tests.

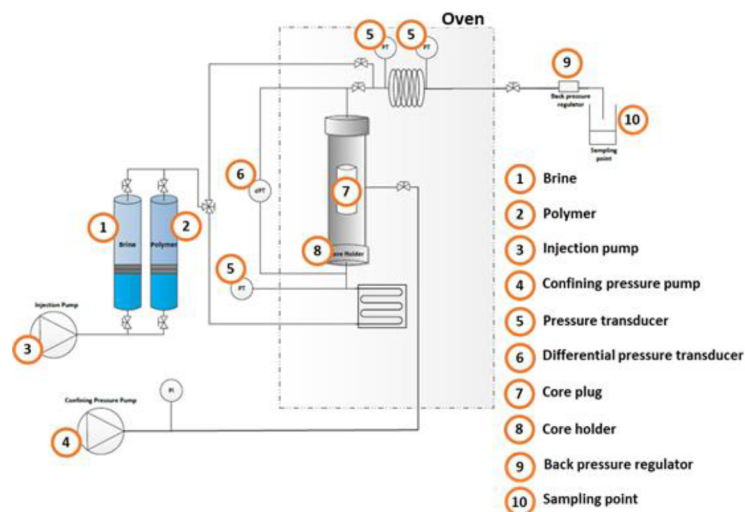


Figure 1—Experimental setup used for single-phase and two-phase core flooding.

Each core plug is loaded into the core holder under a confining pressure of 40 bar and against a back pressure regulator (BPR) set at 5bar(g). CO₂ gas is injected for 30 minutes to flush air from the core and system, and to cool the core plug while enhancing gas dissolution in the subsequent brine injection. Afterward, 8 TH WTP brine is injected until no gas bubbles are visually observed at the outlet. A stable differential pressure (ΔP) further confirms the absence of trapped gas in the core and flow lines.

Brine permeability is measured at three different injection rates at room temperature to establish a baseline. The core is then removed, weighed (while saturated) to estimate porosity using the Archimedes method, and reloaded into the setup. The system is heated to the target temperature, and brine permeability is re-measured at the same injection rates.

The following steps are involved in the injectivity test of a core plug:

- Brine injection at the target temperature to establish a stable AP over multiple pore volumes (PV), used to calculate the resistance factor (RF) and residual resistance factor (RRF).
- Chemical slug injection at 10 ft/day overnight to establish a stable AP over a significant PV for RF calculation.
- Chemical slug injection at various velocities (1-20 ft/day), with effluent samples collected for viscosity measurement.
- Post-brine injection (8 TH WTP) at 10 ft/day overnight until a stable ΔP is reached, used to calculate RRF.

$$RF = \frac{\lambda_{brine}}{\lambda_{AP \text{ solution}}} \quad (1)$$

$$\lambda = \frac{q \times L}{A \times \phi \times \Delta P} \quad (2)$$

$$RRF = \frac{\lambda_{brine \text{ before AP injection}}}{\lambda_{brine \text{ after AP injection}}} \quad (3)$$

Chemical slug injection at varying velocities simulates different shear rates, representing near-wellbore conditions (high velocity >5 ft/day) and deeper reservoir conditions (lower velocity, 1-5 ft/day).

Stability of Chemical Slug: Polymer/AP slugs are non-Newtonian fluids and experience viscosity loss with increasing shear rates. Since shear rate is directly proportional to injection velocity, polymers injected at different velocities undergo mechanical degradation while flowing through porous media. Effluents collected at various injection rates are analyzed to assess viscosity and estimate mechanical stability.

The Mechanical Degradation Rate (MDR) quantifies the percentage viscosity loss compared to the original viscosity of the injected slug:

$$\text{Mechanical Degradation Rate (\%)} = \left(\frac{\eta_{\text{Fresh solution}} - \eta_{\text{Effluent from core}}}{\eta_{\text{Fresh solution}}} \right) * 100 \quad (4)$$

Where η represents the shear viscosity of the solution at a shear rate of 7.944 s^{-1} , at $25^\circ\text{C}/49^\circ\text{C}$.

Effluents collected at lower injection rates (1-5 mL/min) provide insights into slug stability under deep reservoir conditions, while higher rates assess stability near the injection wellbore. Viscosity is measured using a Kinexus Pro+ rheometer across a shear rate range of $1\text{-}20 \text{ s}^{-1}$, with the value at 7.944 s^{-1} used as the reference for MDR calculations.

Two-Phase Core Flooding. *Initialisation of Core Plugs:* The routine core analysis (RCA) procedure is followed, similar to single-phase core flooding. However, longer core plugs (30 cm) are used in two-phase core floods to overcome capillary end effects and minimize dead pore volume. A tracer test is performed on the core plugs to confirm the similarity of the core plugs for comparative analysis.

For the tracer test, a concentration of 60 g/L potassium iodide (KI) is added to 8 TH WTP brine. Effluents are collected in 5 mL fraction tubes, and their densities are measured. Post-tracer brine is also injected to collect additional effluents. The tracer test is conducted at an injection rate of 1 PV/hr at room temperature. After the tracer test, brine permeability is measured at reservoir temperature. Crude oil with a matched viscosity of $20 \text{ mPa}\cdot\text{s}$ is then injected at a lower rate (0.1 mL/min) to determine initial fluid saturations (S_{wi}/S_{oi}), calculated volumetrically from the produced fluids. Initialization is considered complete once no more brine is produced and a stable differential pressure is observed over the injected pore volume.

Effective oil permeability is measured at three injection rates (0.1, 0.2, and 0.1 mL/min). The core plugs are then sealed in the core holder and aged in an oven at 49°C for 8 weeks. After aging, crude oil with matched viscosity is re-injected for 1.5-2 PV to displace aged oil with fresh oil. Effective permeability to oil is re-measured to validate results and detect any plugging caused by fines migration or unwanted particles. Once this process is complete, the core plug is ready for flooding experiments.

Sequence of Fluid Injection: Fluids are injected at an interstitial velocity of 1 ft/day for two-phase core flooding and corresponds to approximately 0.05 mL/min for a core plug with a diameter of 3.8 cm and a length of 30 cm. This low injection rate simulates deep reservoir conditions and provides insights into the mechanical stability of chemical slugs in the presence of crude oil under realistic flow dynamics.

The following injection sequence is adopted for a core flood experiments:

1. Injection of hard brine (8 TH WTP) for 2 pore volumes (PV), continued until oil production ceases and dP stabilizes.
2. Injection of Polymer/AP slug for 2-3 PV, continued until oil production ceases and dP stabilizes.
3. Collection of chemical slug effluents to evaluate mechanical stability.
4. Evaluation of oil recovery factor (ORF) and injectivity (dP).

During the flooding sequence, effluents are collected in 4.5 mL fractional volumes and visually analyzed to calculate the volumetric ORF.

Results and Discussion

Fluid Optimization

Chemical Slug Rheology. Fig. 2 illustrates the impact of alkali addition on a polymer solution viscosity. As expected, the viscosity of the alkali-polymer (AP) slug is slightly higher than that of the polymer-only slug for both polymer products. This increase is primarily attributed to the use of soft brine in preparing the AP slug. It is well-established in the literature that divalent cations such as calcium and magnesium can reduce

polymer viscosity by neutralizing the negative charges on polymer chains (Berry and Fox, 1968; Needham et al., 1974; Tahir et al., 2017). Polymer molecules typically carry negative charges, which generates repulsive forces between chains, resulting in higher viscosity. When divalent cations interact with these charges, they reduce the repulsive forces, leading to a decrease in viscosity. In this study, the concentration of divalent cations in the formation brine is relatively low, so the resulting viscosity loss is minimal.

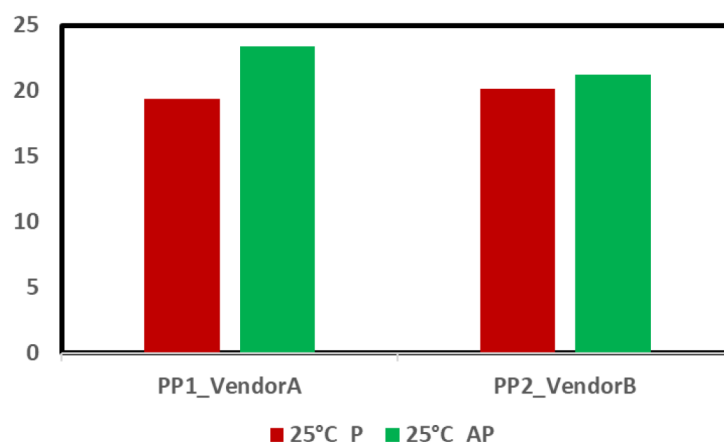


Figure 2—Viscosity comparison of polymer and AP slug for both polymer products at a shear rate of 7.944 s^{-1}

Another contributing factor to the higher viscosity of the AP slug is the increase in pH due to the addition of alkali. A higher pH enhances the negative charge density in the bulk solution, intensifying the repulsive forces among polymer chains. These forces cause the polymer chains to stretch, which manifests as increased viscosity.

Our previous study (Nurmi et al., 2023) demonstrated that the viscosity of the AP slug increases under higher pH conditions due to an aging process in anaerobic environment. This aging effect is further accelerated at elevated temperatures and contrasts with conventional polymer slugs, which typically undergo thermal degradation under similar conditions.

This aging behavior plays a significant role in cost optimization, as the target viscosity of the AP slug can be achieved with lower polymer concentrations. Laboratory tests confirmed that the polymer concentration can be reduced—specifically, PP1 to 450 ppm and PP2 to 200 ppm—while still achieving the desired viscosity under the experimental conditions (Tahir et al., 2024b).

The AP slug also improves injectivity, as lower concentrations are sufficient to reach the in-situ target viscosity when accounting for the aging effect. The selected alkali concentration of 7,500 ppm raises the pH to approximately 10. However, exceeding a pH of 10.5 poses a risk of silica scale precipitation under the target reservoir conditions. Internal testing by chemistry experts concluded that the alkali concentration should not exceed 7,500 ppm to mitigate this risk. Lower concentrations were also deemed unsuitable due to increased alkali adsorption on the rock surface.

Fluid-fluid Interactions. The interfacial tension (IFT) results for the chemical slugs are presented in Fig. 3. Both alkali-polymer (AP) slugs exhibited significantly lower IFT values compared to the base values measured for the polymer slug and the soft 8 TH WTP brine. The initial IFT values for both AP slugs were notably low, in the range of 10^{-3} mN/m , indicating a successful saponification process and effective IFT reduction, consistent with findings in the literature (Arekhov et al., 2020).

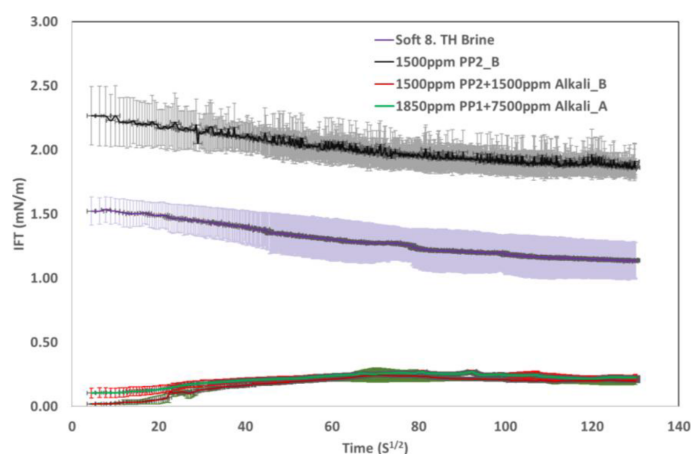


Figure 3—IFT results for AP slugs prepared for both polymer products. Results for PP2_B and soft 8 TH WTP Brine are presented as base cases.

At an initial rotational speed of 3000 rpm, the oil drop appeared elongated. However, as the system was heated, the drop gradually assumed a more spherical shape. To maintain elongation, the rpm was increased incrementally up to 7500rpm. This increase suggests a progressive neutralization of polar oil compounds over time, which led to a gradual rise in IFT. The IFT values shown in Fig. 3 were recorded at 49°C, starting at approximately 10^2 mN/m and increasing slowly, reflecting the ongoing neutralization of polar compounds in the oil phase. Initial contact IFT values are not shown in the figure, as the device began recording only after reaching the target temperature.

These results align with previous studies showing that low-salinity brines are more effective in reducing IFT than hard brines (Farhadi et al., 2021). Since the polymer solution was prepared in hard brine, Fig. 3 confirms that the soft 8 TH WTP brine achieved lower IFT values than the polymer slug. This supports the understanding that commercial HPAM polymers do not significantly contribute to fluid-fluid interactions (Hata and Kasemura, 1980). In both base cases, the oil drop remained spherical at 3000 rpm and did not deform even at higher speeds. Therefore, the Young-Laplace method was used to measure IFT in these cases, as it is suitable for higher IFT values. In contrast, the Vonnegut method was applied to the AP slugs, which had IFT values below 1 mN/m.

Fig. 3 also shows that both AP slugs yielded similar IFT values, attributed to the identical alkali concentration of 7500 ppm. This indicates that the polymer type, despite being sourced from different vendors, had no significant impact on IFT reduction.

The change in emulsion volume for both slug formulations is presented in Fig. 4, with error bars representing the average and standard deviation of the samples. The emulsions formed in the pipettes were nearly equal in volume, mirroring the IFT results. A similar trend in emulsion volume reduction was observed for both polymer products, supporting the hypothesis that polymer type has minimal influence on fluid-fluid interactions, especially when the molecular weights are comparable.

The observed decrease in emulsion volume over time suggests that the emulsions are inherently unstable and prone to phase separation. This observation is consistent with previous findings (Hincapie et al., 2022d, 2022c), although the crude oil used in those studies originated from a different reservoir. Microscopic analysis was conducted to further validate the phase separation and emulsion volume reduction. Unstable emulsions are beneficial for mobilizing the oil phase and can significantly enhance oil recovery. Such emulsions are expected to form under reservoir conditions when AP slugs interact with residual or trapped oil. These emulsions are mobilized by the drag and viscous forces of the injected fluids and are likely to undergo phase separation as they migrate toward the production well. However, their movement may be hindered due to the higher viscosity of stable emulsions compared to crude oil or the AP slug, increasing the risk of emulsion entrapment within the reservoir.

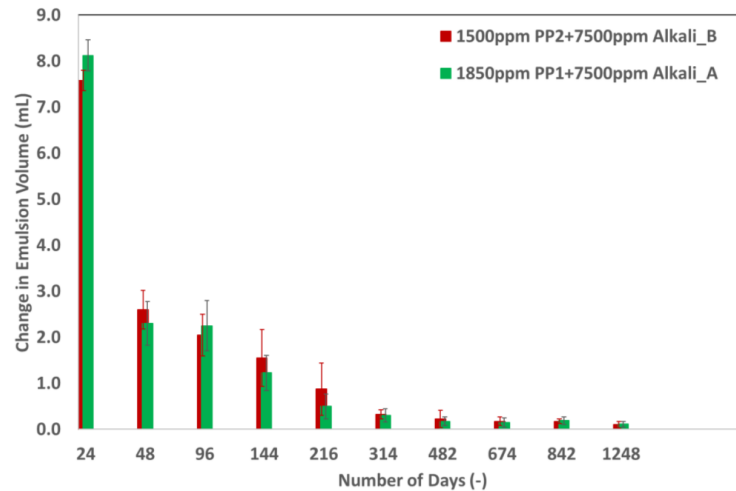


Figure 4—Change in emulsion volume produced by both AP slugs over the number of days at 49°C

Based on the fluid-fluid interaction analysis, AP slugs effectively increase pH and reduce IFT to values in the range of 10^2 to 10^2 mN/m, initiating saponification reactions within the reservoir. The resulting unstable emulsions aid in mobilizing trapped oil, thereby enhancing oil recovery. However, the mechanical stability of AP slugs warrants further investigation.

Rock-Fluid Interactions

Single-Phase Core Floods. *Injectivity Test:* Single-phase injectivity tests were conducted using polymer and alkali-polymer (AP) slugs for two polymer products (PP1 and PP2) (Fig. 5), followed by post-brine flooding with 8 TH WTP brine at 25 °C and an injection velocity of 10 ft/day. As previously mentioned, the polymer concentration used for both the polymer and AP solutions is the same as shown in Fig. 2.

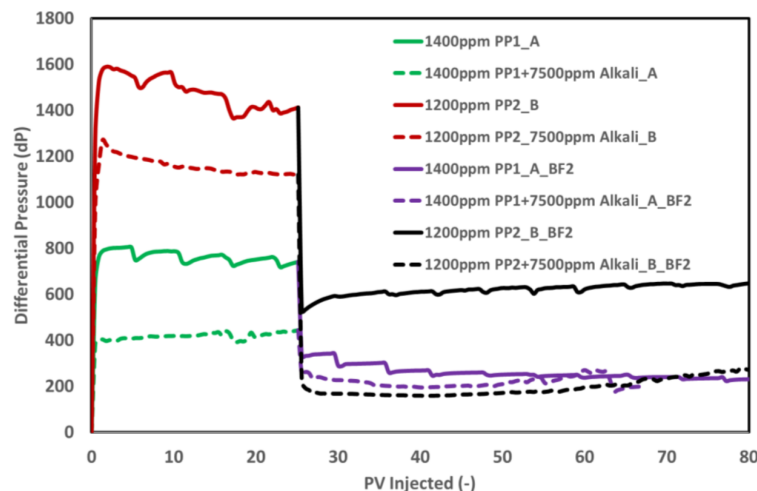


Figure 5—Injectivity tests of polymer and AP slugs for both products at 25°C at injection velocity of 10ft/day

Fig. 5 presents the differential pressure (ΔP) responses:

- For PP1, the polymer slug (solid green line) showed a ΔP approximately 100% higher than the AP slug (dotted green line), despite the AP slug having slightly higher viscosity. This suggests that alkali improves injectivity, likely by reducing polymer adsorption (Al-Murayri et al., 2019).

- Post-brine flooding after the AP slug (dotted purple line) showed lower dP than after the polymer slug (solid purple line), indicating reduced retention. However, dP values converged at higher pore volumes, suggesting similar long-term injectivity.

For PP2:

- The AP slug (dotted red line) exhibited better injectivity than the polymer slug (solid red line).
- However, post-brine flooding after the polymer slug showed significantly higher ΔP , indicating polymer retention or entrapment. In contrast, the AP slug (black dotted line) resulted in much lower ΔP , confirming improved injectivity for post brine flood.

A key observation is that PP2 required significantly higher ΔP than PP1—100% higher for polymer slugs and 200% higher for AP slugs—despite similar viscosities and core permeabilities (~ 300 mD). This is likely due to PP2's higher molecular weight and viscoelasticity, supported by the fact that PP2 achieved the target viscosity at a lower concentration (1200 ppm vs. 1400 ppm for PP1). These findings highlight the importance of viscoelasticity as a screening factor, especially under high injection velocities simulating near-wellbore conditions observed (Elhajjaji et al., 2016; Rock et al., 2020; Tahir et al., 2020, 2017). While reducing PP2 concentration might lower ΔP , it could compromise the mobility ratio under deep reservoir conditions and lower injection rates.

On the effect of injection rate on injectivity and rheology, Fig. 6 shows injectivity results at various injection rates:

- Increasing RF values with injection rate indicate viscoelastic behavior, while decreasing RF suggests shear-thinning.
- PP2-based slugs (polymer and AP) exhibit stronger viscoelasticity than PP1-based slugs, consistent with their molecular weight distribution and lower concentration required for target viscosity.
- At 1 ft/day (deep reservoir conditions), both products showed similar RF values, indicating comparable flow resistance and consistent oil recovery performance (Tahir et al., 2025).
- AP slugs consistently showed better injectivity than polymer-only slugs, despite slightly higher apparent viscosity, reinforcing the role of alkali in enhancing injectivity.

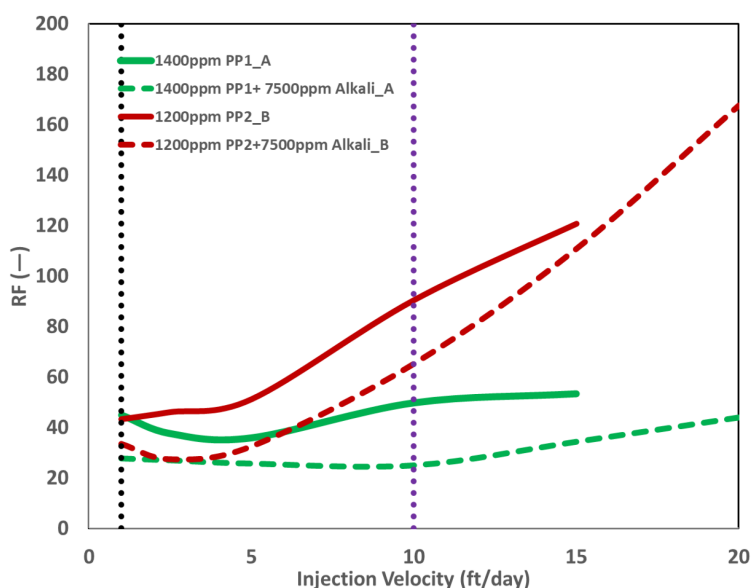


Figure 6—Chemical slugs injectivity at various injection velocities at 25°C

At higher injection rates:

- The PP2-based AP slug showed a steeper RF increase, indicating stronger viscoelastic effects.
- The PP1-based AP slug maintained superior injectivity with minimal shear-thinning or thickening, suggesting more predictable in-situ rheology.

Fig. 6 also shows that AP slugs demonstrate better injectivity than polymer-only slugs for both products. This contradicts the viscosity results in Fig. 2 presenting slightly higher apparent viscosity for AP slugs. Therefore, it can be concluded that alkali significantly enhances polymer injectivity and requires lower wellhead pressure to achieve the defined injection rates. Both AP slugs exhibited similar injectivity up to an injection velocity of 5 ft/day. The viscoelasticity of the PP2-based AP slug causes a steeper increase in RF values at higher injection rates. Consequently, the PP1-based chemical slug shows superior injectivity without pronounced shear-thinning or thickening effects. This suggests that the rheology of the PP1-based AP slug is less influenced by in-situ shear rates and may perform in the reservoir as predicted by laboratory screening.

Injectivity test results at 10 ft/day are summarized in Table 2:

- The PP1-based AP slug demonstrated better injectivity in both high- and low-permeability core plugs compared to PP2-based slugs.
- Higher RF values were observed in low-permeability plugs for both products, as expected.
- AP slugs consistently outperformed polymer-only slugs at similar concentrations.

Table 2—Injectivity results of medium and low perm core plugs

Core Plug (-)	Brine Perm. mD	Temperature (°C)	Chemical Slug (-)	RF (-)	RRF (-)
High Perm	301	25	1500ppm PP1_A	50	15
	328	25	1500ppm PP1 + 7500ppm Alkali_A	25	13
	291	25	1500ppm PP2_B	90	41
	322	25	1500ppm PP2 + 7500ppm Alkali B	65	18
Low Perm	102	49	1500ppm PP2 + 7500ppm Alkali_B	186	3
	106	49	1500ppm PP2 + 7500ppm Alkali_B	178	5
	84	49	1850ppm PP1 + 7500ppm Alkali_A	63	10
	95	49	1850ppm PP1 + 7500ppm Alkali A	51	12

Residual Resistance Factor (RRF) behavior:

- In high-permeability plugs, stable RRF was achieved after injecting 3-5 PV of post-brine flood.
- In low-permeability plugs, stable ΔP was not reached even after 100 PV, indicating delayed stabilization.
- Therefore, transient RRF values in high-permeability plugs provide a more realistic field-scale estimate.

Interestingly, the PP1-based AP slug showed lower RRF in high-permeability plugs but higher RRF in low-permeability plugs, emphasizing the influence of rock permeability on polymer behavior.

Mechanical stability was also assessed:

- For the PP1-based AP slug, RF values were nearly doubled comparing the results of 300 mD at 25 °C to 100 mD at 49 °C (25 vs. 51), indicating good thermal and mechanical stability.
- For the PP2-based AP slug, RF values increased threefold under similar conditions (from 65 to 178), suggesting lower mechanical stability owing to viscoelasticity.

Based on the injectivity test results it is concluded that the presence of alkali significantly enhances the injectivity of alkali-polymer (AP) slugs prepared with polymer products from both vendors. Furthermore, the polymer/AP slugs prepared with PP2 require higher injection pressures and indicate lower injectivity at comparable injection rates. This reduced injectivity is attributed to the stronger viscoelastic properties of PP2 and are supported by its ability to achieve the target viscosity at a lower concentration and its slightly higher molecular weight.

Mechanical Degradation Rate: Mechanical stability of chemical slugs is defined by analyzing the viscosity of effluents produced from core flooding at various injection rates. It also compares the MDR results with injectivity outcomes (RF) to establish a correlation. As shown in Fig. 7, the alkali-polymer (AP) slug exhibited lower MDR values compared to the corresponding polymer-only slug, particularly at lower injection velocities. This suggests that the presence of alkali enhances slug mechanical stability through porous media relative to polymer slugs. The improved mechanical stability of AP slugs contributes to better sweep efficiency in the reservoir.

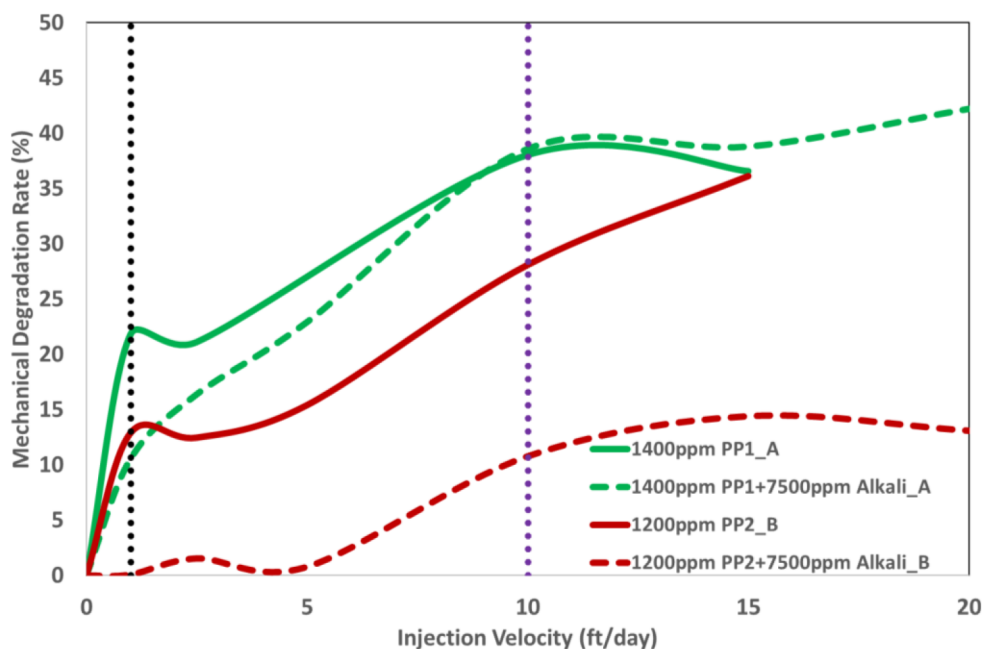


Figure 7—Mechanical stability of chemical slugs as function of increasing injection velocity

Alkali is observed to reduce interfacial tension (IFT) and promote favorable fluid-fluid interactions, which aid in additional oil recovery. However, the enhanced sweep efficiency of the AP slug was not considered an explicit contributor to oil recovery, despite its superior viscosity control and performance. The difference in MDR is especially notable for chemical slugs prepared with PP2. At lower injection velocities (up to 5 ft/day), the MDR for the AP slug is nearly zero, indicating that the effluent viscosity remains unchanged. In contrast, the polymer slug shows an MDR of approximately 15% under the same conditions. Based on core flood results, it is assumed that the PP2-based AP slug undergoes no mechanical degradation under deep reservoir conditions if the permeability is ≥ 300 mD. At higher injection velocities (10 ft/day), the MDR difference remains around 15%, with the polymer slug reaching an MDR of 35%—nearly double that of the AP slug.

Similarly, for PP1-based slugs, the AP slug shows an MDR of around 10%, which is nearly half that of the polymer slug ($\approx 22\%$). This lower MDR is advantageous, as higher effluent viscosity improves sweep efficiency. At an injection velocity of 10 ft/day, both PP1-based slugs exhibit similar MDR values, indicating that the mechanical stability of the PP1-based AP slug is shear-rate dependent and more stable at lower shear rates.

Comparing the AP slugs of both polymer products, the PP2-based AP slug demonstrates superior mechanical stability. The PP1-based AP slug shows a 20% higher viscosity loss at lower shear rates, increasing to 30% at higher injection velocities. As shown in Fig. 5, the injectivity of the PP2-based AP slug requires a pressure drop (ΔP) that is 300% higher than that of the PP1-based AP slug at the same injection rate. Therefore, a technical evaluation is necessary to determine whether it is more feasible to accept a 30% higher viscosity loss or to accommodate a 300% increase in injectivity pressure for chemical slug qualification. The PP1-based AP slug was selected for two core flood screenings to evaluate the mechanical stability of the polymer versus the AP slug, as injectivity is a critical parameter in the polymer qualification process.

Fig. 8 presents the results of MDR for PP2-based polymer and AP slugs at lower permeabilities (100 mD and 200 mD) to support the mechanical stability hypothesis. The AP solution consistently shows lower MDR values compared to the polymer slug, aligning with the trends observed in Fig. 7. At injection velocities of 5 ft/day and 10 ft/day, the polymer slug exhibits nearly 30% higher MDR than the AP slug.

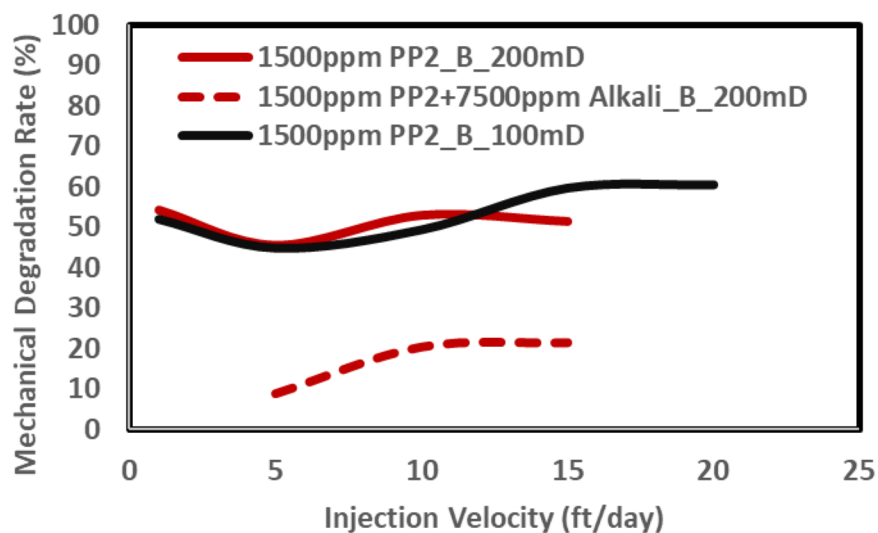


Figure 8—Mechanical stability low low and medium permeability core plug as function injection velocities

Additionally, the MDR for the polymer slug remains nearly constant across both permeability levels, indicating a maximum viscosity loss of approximately 50%. This is further supported by the stable MDR of 50% with increasing injection velocity. Comparing the mechanical stability of the PP2-based AP slug in Fig. 7 and Fig. 8 at 5 ft/day shows that reducing permeability from 300 mD to 200 mD increases the MDR from 0% to 10%, confirming that mechanical stability is permeability (shear rate) dependent.

In conclusion, alkali significantly enhances the mechanical stability of chemical slugs, resulting in lower MDR values. Improved mechanical stability leads to better sweep efficiency and is expected to contribute to additional oil recovery, particularly at lower injection velocities under deep reservoir conditions. The PP2-based AP slug demonstrated 30% better mechanical stability but required a 300% higher injectivity pressure drop compared to the PP1-based AP slug. Therefore, the PP1-based AP slug is selected for further mechanical stability evaluation in two-phase core flood experiments.

Two-Phase Core Floods. As previously mentioned, two-phase core floods are performed using longer core plugs of 30 cm in length. Fig. 9 presents the results of the tracer and post-tracer injections for both core plugs. A higher concentration of potassium iodide (60 g/L) is added to the 8 TH WTP brine as a tracer and density measurements are performed on the produced fractions.

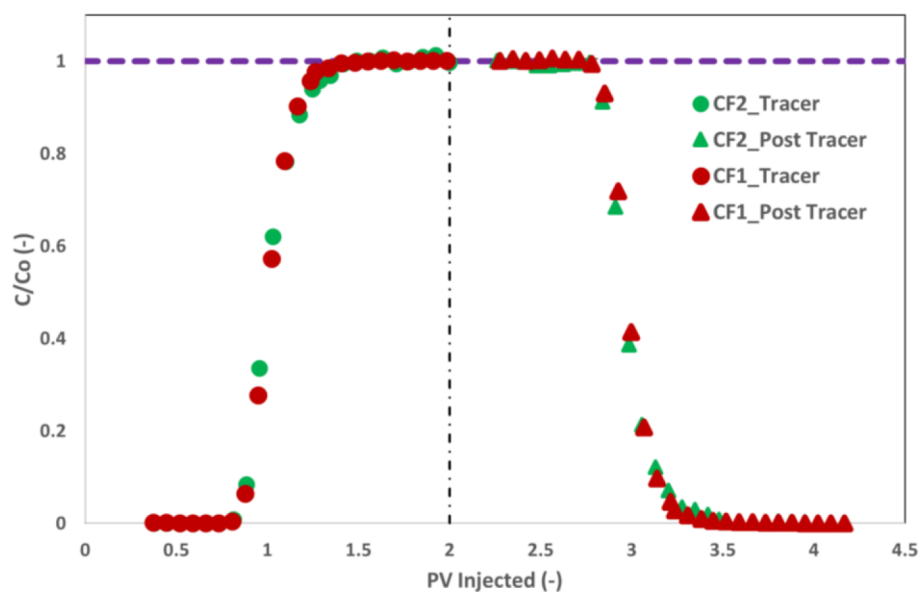


Figure 9—Tracer test results of both core plugs for two phase core flood

An S-shaped breakthrough curve was observed during the tracer test, with $C/C_o = 0.5$ at 1.0 pore volume (PV) injected. This confirms the homogeneity and similarity of both core plugs. The resemblance is further supported by the routine core analysis (RCA) results shown in Table 3. These findings suggest that similar in-situ shear rates can be expected for the polymer and AP slugs under identical injection conditions, especially given the comparable initial and residual oil saturations after the brine flood. The post-tracer trends in Fig. 10 also show good alignment. A value of $C/C_o = 1$ indicates full saturation of the core plug with KI-containing brine, confirming the success of the tracer test. Subsequently, 8 TH WTP brine was injected at 2 PV to displace the KI, resulting in a drop of C/C_o to zero.

After the tracer test, the core plugs were initialized, and S_{oi}/S_{wi} saturations were established as shown in Table 3. Primary brine floods were conducted to simulate field conditions and establish remaining oil saturation. A good correlation between oil production and pressure drop during the brine flood is observed in Table 3 and Fig. 10.

Table 3—RCA and oil recovery for both core plugs used for polymer/AP slugs injection

Core Plug	Length cm	Diameter cm	Poro .(%)	Perm. (B) mD	PV mL	Soi (-)	ORF-Brine (%)	Additional ORF (Slug (%))
1	29.8	3.81	21.2	277	69.81	0.66	38.7	2.20
2	29.8	3.81	20.44	284	69.4	0.68	38.9	23.27

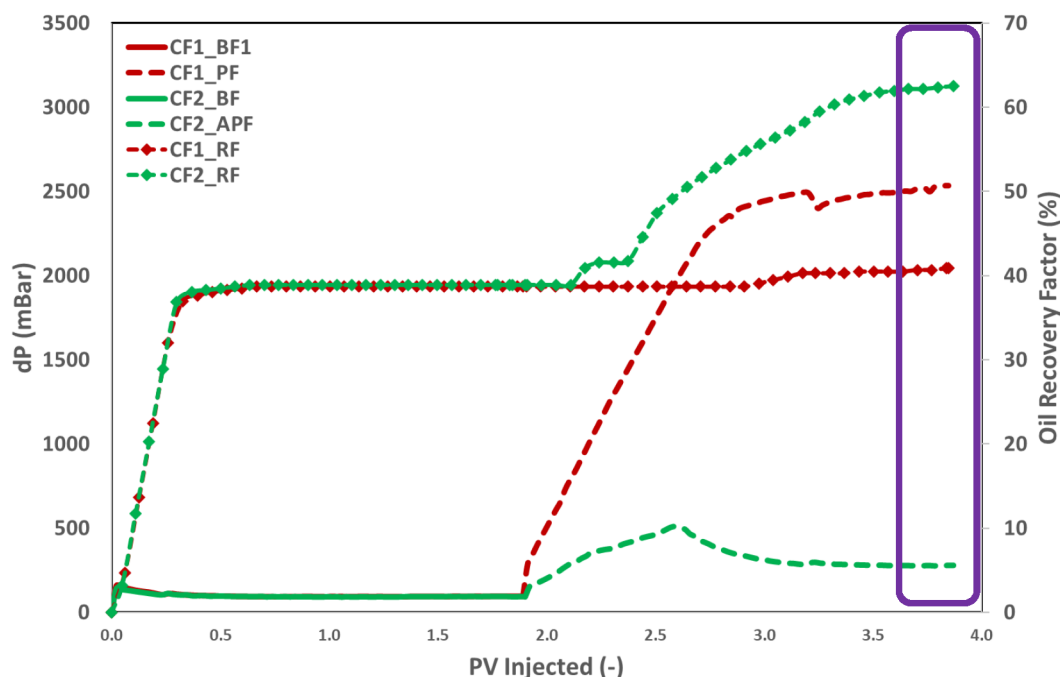


Figure 10—Two-phase core flood for polymer and AP slug to compare injectivity and oil recovery factor

A 1500 ppm polymer flood (PP1) was injected into the first core plug (CF1), while an alkali-polymer (AP) slug containing 1500 ppm polymer and 7500 ppm alkali was injected into the second core plug (CF2). A significantly higher ΔP was observed for the polymer slug, consistent with single-phase core flood results. However, the peak ΔP for the AP slug was five times lower, indicating improved injectivity. This is notable given that the AP slug had slightly higher viscosity and the in-situ shear rate was expected to be similar at an interstitial injection velocity of 1 ft/day. In the single-phase core flood (Table 2), the AP slug showed a twofold improvement in injectivity. In the two-phase flood, the presence of residual oil enhanced this improvement to fivefold, suggesting that fluid-fluid (f-f) interactions contributed to the improved injectivity.

The AP slug achieved a 23% increase in additional oil recovery, while the polymer slug contributed only 2%, as shown in Table 3. One key factor behind the higher recovery from the AP slug is the f-f interactions induced by alkali. Additionally, Fig. 11 highlights that mechanical stability of the chemical slug also played a significant role in the enhanced oil recovery.

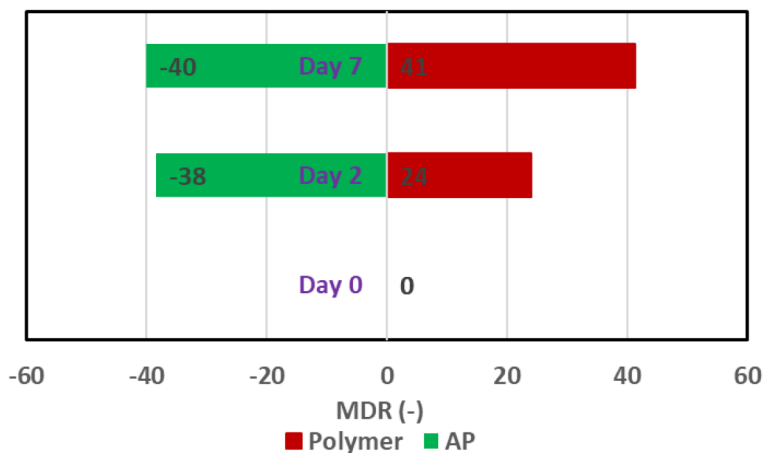


Figure 11—Mechanical degradation rate for polymer/AP slug effluents produced from two phase core flood

Day 0 marks the preparation and injection of the chemical slugs. The viscosity of the freshly prepared slug was used as the reference, with an initial MDR of zero. Breakthrough occurred within 24 hours, and effluents were collected in 4.5 mL fractions. After 48 hours and 2 PV of injection, stabilization in ΔP and oil recovery was achieved. The final effluent samples were used to measure viscosity under the same conditions as the reference, highlighted by the purple rectangle in Fig. 10.

A 24% loss in polymer viscosity was observed on Day 2 indicating significant degradation of the polymer under in-situ shear rate conditions (Real, 2023). In contrast, the AP slug showed an unexpected 38% increase in viscosity. According to the concept of mechanical stability, the AP slug should retain or slightly reduce its viscosity. This increase is likely due to a combination of aging under high pH and alkali adsorption on the rock surface, which acts as a sacrificial agent preserving polymer viscosity. This aging effect has been previously demonstrated (Hincapie et al., 2022b, 2022a; Nurmi et al., 2023), and is known to be time-dependent and accelerated by higher temperatures. Notably, those studies were conducted without crude oil.

It is assumed the presence of crude oil or the saponification process probably enhance the aging effect, resulting in a significant increase in AP slug viscosity. On Day 7, viscosity measurements from stored effluent samples showed a further decrease to 41% for the polymer slug, likely due to room temperature storage and oxygen exposure (Real, 2023). In contrast, the AP slug viscosity increased to 40%. A thin oil film observed on the AP sample may have protected it from oxidative degradation. This 40% viscosity increase under ambient conditions supports the mechanical and chemical stability of the AP slug. Therefore, the significant additional oil recovery observed with the AP slug is attributed not only to f-f interactions but also to its superior mechanical stability. The viscosity results in Fig. 11 suggest that aging tests should be conducted in the presence of crude oil to confirm the observed viscosity gains.

Additional Benefits and Challenges of an AP slug

This study confirms that the alkali-polymer (AP) slug significantly improves chemical slug injectivity, aligning well with previous findings (Hincapie et al., 2022b; Nurmi et al., 2023; Tahir et al., 2024a). Furthermore, the enhancement in mechanical stability is also supported by the outcomes of this study. The combination of better injectivity, improved mechanical/chemical stability and in-situ development of f-f interactions contributes to significantly higher oil recovery compared to a polymer slug (Hincapie et al., 2022c; Hoffmann et al., 2024; Tahir et al., 2024a).

One of the key mechanisms behind this performance is the aging process of the AP slug, driven by high pH conditions and increased polymer anionicity. This process promotes the development of repulsive forces among polymer chains, which increases viscosity over time (Nurmi et al., 2023; Tahir et al., 2024b). As a result, lower initial polymer concentrations can be used to achieve the target viscosity, which not only reduces chemical costs but also improves injectivity due to the lower viscosity of the freshly prepared slug (Juárez et al., 2020). This cost optimization—achieved through reduced polymer usage, improved injectivity, and enhanced oil recovery—makes the AP slug a multifunctional and economically attractive candidate for field-scale cEOR deployment.

Despite its advantages, the AP slug also presents several technical and operational challenges, which must be carefully managed:

1. **Emulsion Stability and Injectivity Risks:** Alkali alone can form stable emulsions with crude oil, which may increase viscosity and hinder injectivity, potentially compromising the success of EOR operations (Hunter et al., 2013; Izadi et al., 2018; McInnis et al., 2013; Qing et al., 2002). However, the use of an AP slug mitigates this risk by forming unstable emulsions that undergo phase separation after mobilization within the reservoir (Hincapie et al., 2022c; Tahir et al., 2024a).
2. **Scaling and Precipitation Risks:** The use of aggressive alkalis such as NaOH increases the risk of scale precipitation (e.g., CaCO_3 , Mg(OH)_2) in production wells, which can lead to equipment fouling and

pilot failure (McInnis et al., 2013; Sheng, 2017). To mitigate this, milder alkalis like sodium carbonate (Na_2CO_3) are preferred due to their better pH control and lower scaling potential.

3. **Water Softening Requirements:** Na_2CO_3 is incompatible with divalent cations such as Ca^{2+} and Mg^{2+} . Therefore, a water softening unit is required to remove these ions from injection water, both for the preflush and for AP slug preparation (Marx et al., 2023). While this adds to the capital and operational costs, in our case, the low concentrations of Ca^{2+} and Mg^{2+} in produced water make the cost of water softening negligible compared to the benefits gained—namely:
 - Reduced polymer concentration,
 - Improved mechanical stability, and
 - Increased oil recovery.
4. **Field-Specific Economic and Risk Assessments:** Given the complexity of AP slug behavior and its sensitivity to reservoir conditions, it is essential to conduct comprehensive economic screening and risk assessments before field implementation (Janczak et al., 2023). These should include:
 - Compatibility testing with formation water and crude oil,
 - Scaling potential analysis,
 - Long-term injectivity monitoring, and
 - Evaluation of surface facility requirements.

The AP slug offers various advantages in cEOR applications by combining improved injectivity, enhanced mechanical and chemical stability, and cost-effective viscosity control through aging. While challenges such as emulsion stability, scaling, and water softening must be addressed, these are manageable with proper design and operational strategies. Overall, the AP slug emerges as a technically robust and economically viable solution for improving oil recovery in suitable reservoir conditions.

Conclusions

This study provides compelling evidence that alkali addition in polymer slugs significantly enhances both injectivity and mechanical stability, offering a robust and cost-effective solution for chemical enhanced oil recovery (cEOR) in heterogeneous reservoirs. The key conclusions are:

- **Enhanced Injectivity:** AP slugs consistently demonstrated superior injectivity compared to polymer-only slugs across both single-phase and two-phase core floods. This improvement is attributed to reduced polymer adsorption and favorable rock surface modifications induced by alkali.
- **Improved Mechanical Stability:** The AP slugs exhibited significantly lower mechanical degradation rates (MDR), particularly under low shear conditions representative of deep reservoir environments. In contrast, polymer-only slugs suffered up to 41% viscosity loss, while AP slugs showed up to 40% viscosity gain due to in-situ aging.
- **In-Situ Aging and Viscosity Gain:** The AP slug's viscosity increased over time due to high pH-induced polymer chain repulsion and alkali adsorption on rock surfaces. This aging effect enables the use of lower polymer concentrations while maintaining or even enhancing viscosity during propagation, thus optimizing chemical costs.
- **Additional Oil Recovery:** Two-phase core floods confirmed that AP slugs achieved up to 23% additional oil recovery, compared to only 2% from polymer slugs. This is attributed to a

combination of improved sweep efficiency, enhanced mechanical stability, and in-situ fluid-fluid interactions (e.g. saponification and emulsion mobilization).

- **Polymer Product Selection is key:** While both PP1 and PP2-based AP slugs benefited from alkali addition, PP1-based slugs offered better injectivity and more predictable rheological behavior under varying shear rates. PP2-based slugs, although mechanically more stable, required significantly higher injection pressures due to their stronger viscoelasticity.
- **Field-Relevant Validation:** The study's methodology—using realistic injection velocities, reservoir-matched fluids, and long core plugs—ensures that the findings are directly applicable to field-scale implementation. Tracer tests and RCA confirmed the homogeneity of core plugs, validating the comparative analysis.
- **Operational and Economic Implications:** The ability to reduce polymer concentration without compromising performance, combined with improved injectivity and oil recovery, positions the AP slug as a technically and economically viable EOR solution. However, field deployment must consider water softening requirements, emulsion stability, and scaling risks.

Acknowledgments

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Abbreviations

AP	Alkali-polymer
cEOR	chemical-enhanced oil recovery
HPAM	Hydrolyzed polyacrylamide
IFT	Interfacial tension
mD	Milli darcy
MDa	Million Daltons
PV	Pore volume
TAN	Total acid number
TH	Torton Horizon
WTP	Water treatment plant
SARA	Saturate, aromatic, resin and asphaltene
RF	Resistance factor
RRF	Residual resistance factor
ORF	Oil Recovery factor
MDR	Mechanical Degradation Rate
PP1	Polymer product 1
PP2	Polymer product 2
ΔP	Pressure drop across core plug

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